We create chemistry

# Safety data sheet



## 1. Identification

### Product identifier

## Herbadox 400 EC

### Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: crop protection product, herbicide

### Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Crop Protection

Telephone: +49 621 60-27777
E-mail address: Produktinformation-Pflanzenschutz@basf.com

### Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

## 2. Hazards Identification

### Classification of the substance or mixture

According to UN GHS criteria

Asp. Tox. 1
Acute Tox. 5 (oral)
Skin Corr./Irrit. 3
Eye Dam./Irrit. 2A

Aquatic Acute 1
Aquatic Chronic 1

For the classifications not written out in full in this section the full text can be found in section 16.

## Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Danger

Hazard Statement:

| | |
|---|---|
| H319 | Causes serious eye irritation. |
| H316 | Causes mild skin irritation. |
| H303 | May be harmful if swallowed. |
| H304 | May be fatal if swallowed and enters airways. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |

Precautionary Statement:

| | |
|---|---|
| P101 | If medical advice is needed, have product container or label at hand. |
| P102 | Keep out of reach of children. |
| P103 | Read carefully and follow all instructions. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P280 | Wear eye protection. |
| P264 | Wash contaminated body parts thoroughly after handling. |

Precautionary Statements (Response):

| | |
|---|---|
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P301 + P310 | IF SWALLOWED: Immediately call a POISON CENTER or physician. |
| P391 | Collect spillage. |
| P331 | Do NOT induce vomiting. |
| P337 + P313 | If eye irritation persists: Get medical attention. |
| P332 + P313 | If skin irritation occurs: Get medical attention. |

Precautionary Statements (Storage):

| | |
|---|---|
| P405 | Store locked up. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Labeling of special preparations (GHS):
May produce an allergic reaction. Contains: N-(1-Ethylpropyl)-2,6-dinitro-3,4-xylidine

According to UN GHS criteria

Hazard determining component(s) for labelling: pendimethalin (ISO); N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine, Solvent naphtha (petroleum), heavy arom., 2-Ethylhexan-1-ol

### Other hazards

According to UN GHS criteria

See section 12 - Results of PBT and vPvB assessment.

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

## 3. Composition/Information on Ingredients

### Substances

Not applicable

### Mixtures

Chemical nature

crop protection product, herbicide

Hazardous ingredients (GHS)
According to UN GHS criteria

pendimethalin (ISO); N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine

| | |
|---|---|
| Content (W/W): 37,67 % | Acute Tox. 5 (oral) |
| CAS Number: 40487-42-1 | Skin Sens. 1B |
| EC-Number: 254-938-2 | Aquatic Acute 1 |
| INDEX-Number: 609-042-00-X | Aquatic Chronic 1 |
| | M-factor acute: 100 |
| | M-factor chronic: 10 |
| | H303, H317, H400, H410 |

Solvent naphtha (petroleum), heavy arom.

| | |
|---|---|
| Content (W/W): < 60 % | Asp. Tox. 1 |
| CAS Number: 64742-94-5 | Aquatic Acute 2 |
| EC-Number: 265-198-5 | Aquatic Chronic 2 |
| INDEX-Number: 649-424-00-3 | H304, H401, H411 |

Oxirane, methyl-, polymer with oxirane, mono(nonylphenyl) ether, branched

| | |
|---|---|
| Content (W/W): < 5 % | Aquatic Chronic 4 |
| CAS Number: 68891-11-2 | H413 |

2-Ethylhexan-1-ol

| | |
|---|---|
| Content (W/W): < 5 %<br>CAS Number: 104-76-7<br>EC-Number: 203-234-3 | Flam. Liq. 4<br>Acute Tox. 5 (oral)<br>Acute Tox. 4 (Inhalation - mist)<br>Skin Corr./Irrit. 2<br>Eye Dam./Irrit. 2A<br>STOT SE 3 (irr. to respiratory syst.)<br>Aquatic Acute 3<br>H227, H319, H315, H332, H303, H335, H402 |

For the classifications not written out in full in this section the full text can be found in section 16.

## 4. First-Aid Measures

### Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention. Do not induce vomiting due to aspiration hazard.

### Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

### Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## 5. Fire-Fighting Measures

### Extinguishing media

Suitable extinguishing media:
water spray, dry powder, foam, carbon dioxide

### Special hazards arising from the substance or mixture

Carbon monoxide, Carbon dioxide, Hydrogen chloride, nitrogen oxides, sulfur oxides, halogenated compounds

The substances/groups of substances mentioned can be released in case of fire.

### Advice for fire-fighters

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Keep containers cool by spraying with water if exposed to fire. In case of fire and/or explosion do not breathe fumes. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

## 6. Accidental Release Measures

### Personal precautions, protective equipment and emergency procedures

Do not breathe vapour/spray. Use personal protective clothing. Avoid contact with the skin, eyes and clothing.

### Environmental precautions

Do not discharge into the subsoil/soil. Do not discharge into drains/surface waters/groundwater.

### Methods and material for containment and cleaning up

For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations.

## 7. Handling and Storage

### Precautions for safe handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
Vapours may form ignitable mixture with air. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

### Conditions for safe storage, including any incompatibilities

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Storage stability:
Storage duration: 24 Months

### Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

## 8. Exposure Controls/Personal Protection

### Control parameters

Components with occupational exposure limits

104-76-7: 2-Ethylhexan-1-ol
64742-94-5: Solvent naphtha (petroleum), heavy arom.

### Exposure controls

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic, alkaline compounds and toxic particles (e. g. EN 14387 Type ABEK-P3)

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

## 9. Physical and Chemical Properties

### Information on basic physical and chemical properties

| | |
|---|---|
| Form: | liquid |
| Colour: | yellow to orange |
| Odour: | characteristic |
| Odour threshold: | Not determined due to potential health hazard by inhalation. |
| pH value: | The product has not been tested. |
| Melting point: | < -5 °C<br>Information applies to the solvent. |
| Boiling point: | approx. 230 - 300 °C<br>Information applies to the solvent. |

| | |
|---|---|
| Flash point: | approx. 100 °C<br>Information applies to the solvent. |
| Evaporation rate: | not applicable |
| Flammability: | not highly flammable |
| Lower explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. |
| Upper explosion limit: | As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use. |
| Ignition temperature: | > 400 °C<br>Information applies to the solvent. |
| Vapour pressure: | approx. 0,02 - 0,03 hPa<br>(20 °C) |
| Density: | approx. 1,06 g/cm3<br>(20 °C) |
| Relative vapour density (air): | not applicable |
| Solubility in water: | emulsifiable<br>(20 °C) |
| Partitioning coefficient n-octanol/water (log Kow): | not applicable |
| Thermal decomposition: | No decomposition if stored and handled as prescribed/indicated. |
| Viscosity, dynamic: | approx. 2,7 - 5,8 mPa.s<br>Information applies to the solvent. |
| Explosion hazard: | Based on the chemical structure there is no indication of explosive properties. |
| Fire promoting properties: | Based on its structural properties the product is not classified as oxidizing. |

### Other information

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

## 10. Stability and Reactivity

### Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

### Chemical stability
The product is stable if stored and handled as prescribed/indicated.

### Possibility of hazardous reactions
No hazardous reactions if stored and handled as prescribed/indicated.

### Conditions to avoid
See SDS section 7 - Handling and storage.

### Incompatible materials

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

### Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Of low toxicity after single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
LD50 rat (oral): 3.821 mg/kg

LC50 rat (by inhalation): > 5,35 mg/l 4 h

LD50 rat (dermal): > 2.000 mg/kg
No mortality was observed.

Irritation

Assessment of irritating effects:
Skin contact causes slight irritation. Eye contact causes irritation. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
Skin corrosion/irritation rabbit: Slightly irritating.

Serious eye damage/irritation rabbit: Irritant.

Respiratory/Skin sensitization

Assessment of sensitization:
There is no evidence of a skin-sensitizing potential. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
modified Buehler test guinea pig: Non-sensitizing.

Germ cell mutagenicity

Assessment of mutagenicity:
Mutagenicity tests revealed no genotoxic potential. The product has not been tested. The statement has been derived from the properties of the individual components.

Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: pendimethalin (ISO); N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine*
*Assessment of carcinogenicity:*
*In long-term studies in rats the substance induced thyroid tumors. The effect is caused by an animal specific mechanism that has no human counter part. In long-term studies in mice in which the substance was given by feed, a carcinogenic effect was not observed.*
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Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect. The product has not been tested. The statement has been derived from the properties of the individual components.

Developmental toxicity

Assessment of teratogenicity:
Animal studies gave no indication of a developmental toxic effect at doses that were not toxic to the parental animals. The product has not been tested. The statement has been derived from the properties of the individual components.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: pendimethalin (ISO); N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine*
*Assessment of repeated dose toxicity:*
*No substance-specific organtoxicity was observed after repeated administration to animals. Adaptive effects were observed after repeated exposure in animal studies.*

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Aspiration hazard

May also damage the lung at swallowing (aspiration hazard).
The product has not been tested. The statement has been derived from the properties of the individual components.

Other relevant toxicity information

Misuse can be harmful to health.

## 12. Ecological Information

### Toxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: pendimethalin (ISO); N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine*
*Toxicity to fish:*
*LC50 (96 h) 0,196 mg/l, Oncorhynchus mykiss*
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*Information on: pendimethalin (ISO); N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine*
*Aquatic invertebrates:*
*EC50 (48 h) 0,147 mg/l, Daphnia magna*
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*Information on: pendimethalin (ISO); N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine*
*Aquatic plants:*
*EC50 (7 d) 0,012 mg/l, Lemna gibba*

*EC10 (7 d) 0,0029 mg/l, Lemna gibba*

*No observed effect concentration (7 d) 0,003 mg/l, Lemna gibba*

*EC50 (72 h) 0,0093 mg/l, Selenastrum capricornutum*

*EC10 (72 h) 0,0028 mg/l, Selenastrum capricornutum*
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*Information on: pendimethalin (ISO); N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine*
*Chronic toxicity to fish:*
*No observed effect concentration (179 d) 0,02 mg/l, Brachydanio rerio (semistatic)*

*No observed effect concentration (288 d) 0,006 mg/l, Pimephales promelas*
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*Information on: pendimethalin (ISO); N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine*
*Chronic toxicity to aquatic invertebrates:*

*No observed effect concentration (21 d) 0,0145 mg/l, Daphnia magna*
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## Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: pendimethalin (ISO); N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine*
*Assessment biodegradation and elimination (H2O):*
*Not readily biodegradable (by OECD criteria).*
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## Bioaccumulative potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: pendimethalin (ISO); N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine*
*Bioaccumulation potential:*
*Bioconcentration factor: 3.300*
*Based on a weight of evidence, the compound will not bioaccumulate.*
---

## Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: pendimethalin (ISO); N-(1-ethylpropyl)-2,6-dinitro-3,4-xylidine*
*Assessment transport between environmental compartments:*
*Volatility: The substance will slowly evaporate into the atmosphere from the water surface.*
*Adsorption in soil: Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.*
---

## Results of PBT and vPvB assessment

The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

## Other adverse effects

The product does not contain substances that are listed in the Montreal Protocol on substances that deplete the ozone layer.

## Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

## 13. Disposal Considerations

### Waste treatment methods

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

## 14. Transport Information

**Land transport**

ADR

| | |
|---|---|
| UN number | UN3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains PENDIMETHALIN, SOLVENT NAPHTHA) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

RID

| | |
|---|---|
| UN number | UN3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains PENDIMETHALIN, SOLVENT NAPHTHA) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| UN number | UN3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains PENDIMETHALIN, SOLVENT NAPHTHA) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

Transport in inland waterway vessel

Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| UN number: | UN 3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains PENDIMETHALIN, SOLVENT NAPHTHA) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes<br>Marine pollutant: YES |
| Special precautions for user: | None known |

**Air transport**

IATA/ICAO

| | |
|---|---|
| UN number: | UN 3082 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (contains PENDIMETHALIN, SOLVENT NAPHTHA) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

### Transport in bulk according to Annex II of MARPOL and the IBC Code

| | |
|---|---|
| Regulation: | Not evaluated |
| Shipment approved: | Not evaluated |
| Pollution name: | Not evaluated |
| Pollution category: | Not evaluated |
| Ship Type: | Not evaluated |

**Further information**
Product may be shipped as non-hazardous in suitable packages containing a net quantity of 5 L or less under the provisions of various regulatory agencies: ADR, RID, ADN: Special Provision 375; IMDG: 2.10.2.7; IATA: A197; TDG: Special Provision 99(2); 49CFR: §171.4 (c) (2) and also the Special Provision 375 in Appendix B which is regulated in China "Regulations Concerning Road Transportation of Dangerous Goods Part 3: Index of dangerous goods name and transportation requirements" (JT/T 617.3)

## 15. Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

To avoid risks to man and the environment, comply with the instructions for use.

## 16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

| | |
|---|---|
| Asp. Tox. | Aspiration hazard |
| Acute Tox. | Acute toxicity |
| Skin Corr./Irrit. | Skin corrosion/irritation |
| Eye Dam./Irrit. | Serious eye damage/eye irritation |
| Aquatic Acute | Hazardous to the aquatic environment - acute |
| Aquatic Chronic | Hazardous to the aquatic environment - chronic |
| Skin Sens. | Skin sensitization |
| Flam. Liq. | Flammable liquids |
| STOT SE | Specific target organ toxicity — single exposure |
| H303 | May be harmful if swallowed. |
| H317 | May cause an allergic skin reaction. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |
| H304 | May be fatal if swallowed and enters airways. |
| H401 | Toxic to aquatic life. |
| H411 | Toxic to aquatic life with long lasting effects. |
| H413 | May cause long lasting harmful effects to aquatic life. |
| H227 | Combustible liquid. |
| H319 | Causes serious eye irritation. |
| H315 | Causes skin irritation. |
| H332 | Harmful if inhaled. |
| H335 | May cause respiratory irritation. |
| H402 | Harmful to aquatic life. |

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.